

DETAILED DESCRIPTION OF THE INVENTION — FUTURE SENSOR EXPANSION AND TECHNOLOGY-AGNOSTIC IMPLEMENTATION VARIANTS

The intraluminal physiologic monitoring device may be implemented using a wide range of sensing technologies, including technologies not yet developed at the time of filing. The invention therefore encompasses technology-agnostic sensor integration frameworks enabling future expansion without departing from the scope of the disclosed system.

In various embodiments, sensing systems may include pressure sensors, impedance sensors, optical sensors, ultrasound transducers, time-of-flight ranging systems, temperature sensors, biochemical sensors, motion sensors, capacitive sensors, electromagnetic sensing elements, or combinations thereof. Sensor placement, quantity, and configuration may vary depending on intended application.

Future embodiments may incorporate emerging sensing modalities including molecular detection sensors, advanced imaging systems, microfluidic analyzers, biosignal acquisition interfaces, or nanoscale sensing platforms capable of measuring physiologic characteristics beyond current technological capabilities.

Sensor modules may be distributed along the device body, embedded within compliant structures, integrated into inflatable regions, positioned within external pull tab assemblies, or deployed as detachable modules connected through modular interfaces.

Technology-agnostic architecture may permit substitution of alternative sensing technologies performing equivalent measurement functions. Replacement of specific sensor types shall not be considered a departure from the invention when functional objectives remain consistent with physiologic monitoring.

The device may further integrate software-defined sensing frameworks allowing sensor behavior, sampling strategy, or analytic interpretation to evolve through firmware updates or computational improvements independent of hardware modification.

Certain embodiments may enable dynamic sensor activation based on detected physiologic states, conserving energy while maintaining measurement capability. Adaptive sensing strategies may respond to contextual or environmental conditions.

Future implementations may combine multiple sensing technologies into unified sensing arrays capable of generating high-dimensional physiologic representations derived from multi-modal data fusion.

The future sensor expansion and technology-agnostic implementation concepts described herein are illustrative rather than limiting. Any sensing technology capable of acquiring physiologic information within an intraluminal monitoring context is considered within the scope of the present disclosure.